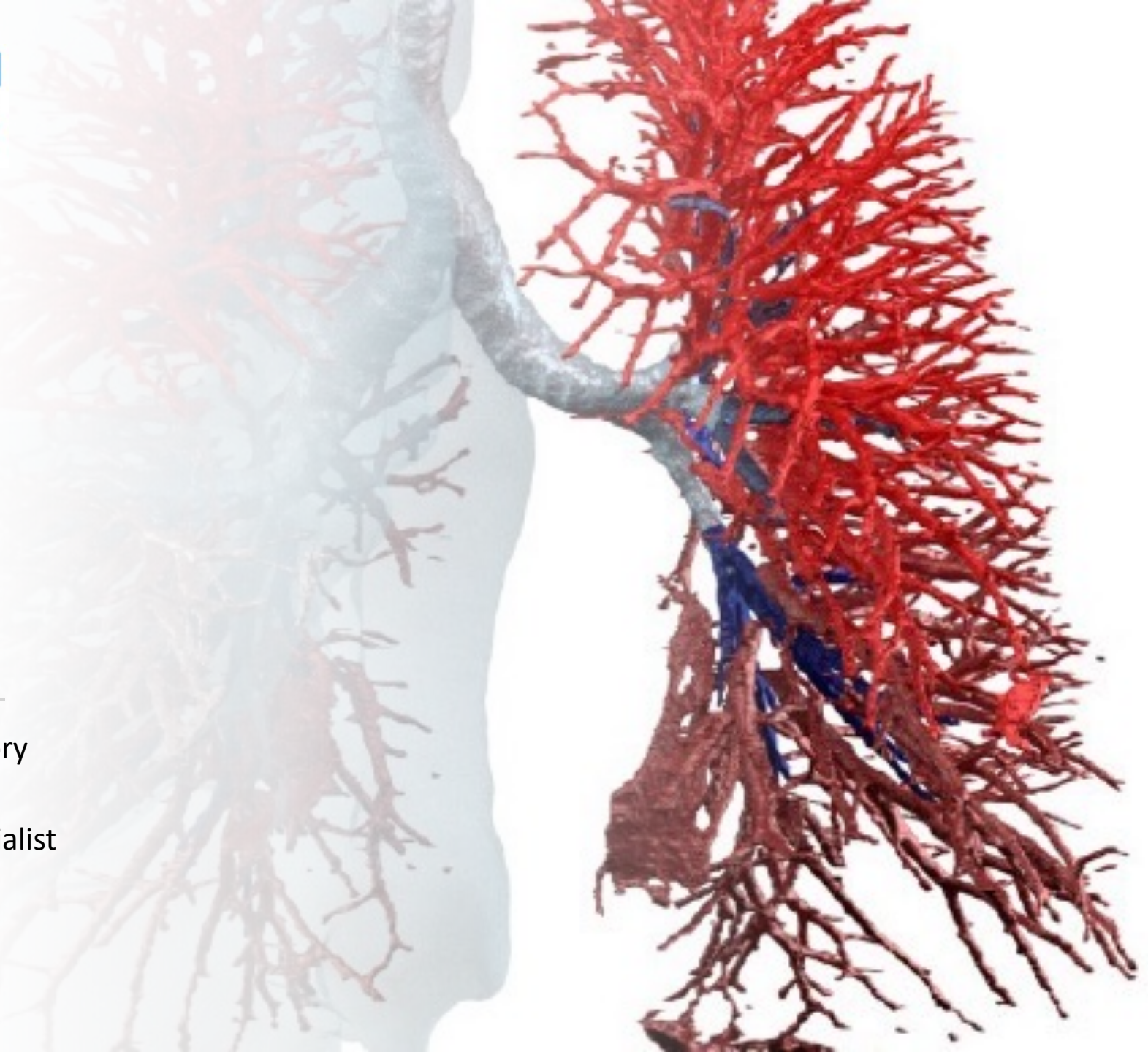
Guy's and St Thomas' NHS Foundation Trust

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Ventilation - Perfusion & Lung Mechanics



Beccy Hall - Highly Specialist Cardiorespiratory Physiotherapist

With thanks to Ashleigh Thorn - Highly Specialist Respiratory Physiotherapist

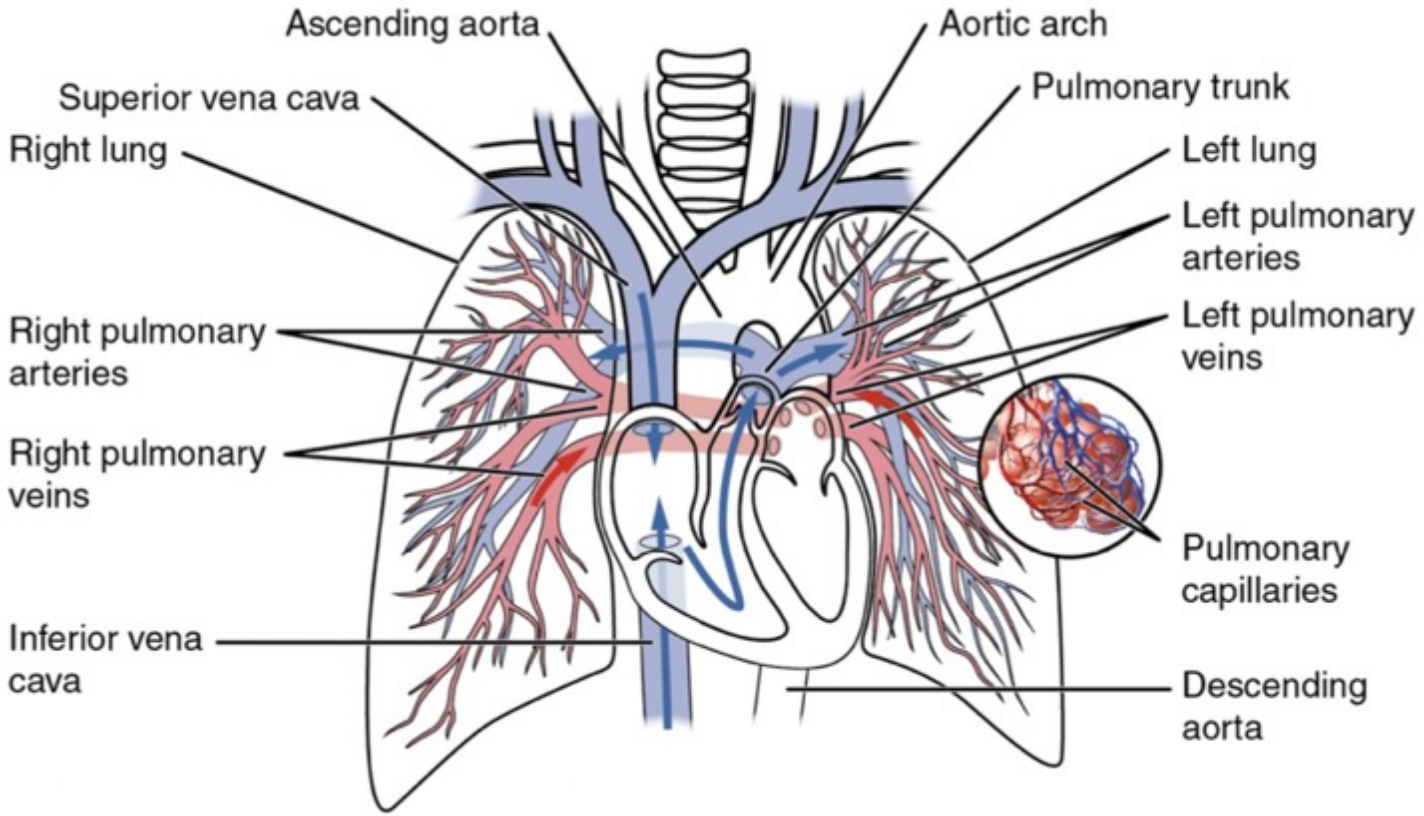
LONDON METROPOLITAN UNIVERSITY

Learning Objectives:

By the end of this lecture, you should have a greater understanding of:

- The properties of the pulmonary circulation.
- Comparison of pulmonary and systemic circulation
- Ventilation and perfusion in the healthy lung
- The importance of matching ventilation and perfusion in different areas of the lung
- The effects of having areas of high and low ventilation / perfusion ratios (ventilation-perfusion mismatching).
- How maintencance of V/Q match is achieved
- Understand the causes of impaired oxygen usage by tissues
- Mechanics of ventilation
- Physical factors affecting ventilation

Pulmonary Circulation:

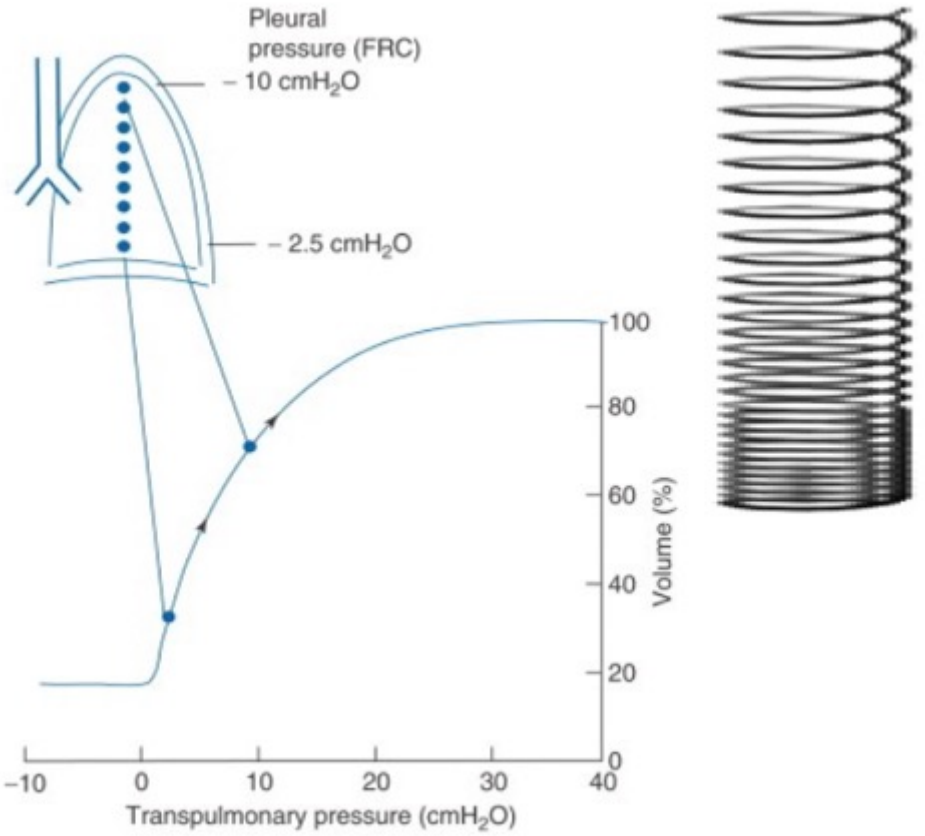


Pulmonary vs Systemic Circulation

Category	Pulmonary Circulation	Systemic Circulation
Function	Moves blood between heart and lungs	Regulates blood supply to various organs
Gas exchange functions	Absorption alveolar O2 Released of capillary CO2	Release of capillary O2 Absorption of CO2
Blood pressure	Low	High
Anatomy	Thin vessels Minimal smooth muscle	Thick vessels Abundant thick smooth muscle
Circulatory Regulation	Minimal capacity to regulate flow	Regional blood flow regulation
Regional distribution of blood flow	Blood flow affected by - gravity - alveolar recruitment - hypoxic vasoconstriction	Significant variation of organ-specific regional blood flow

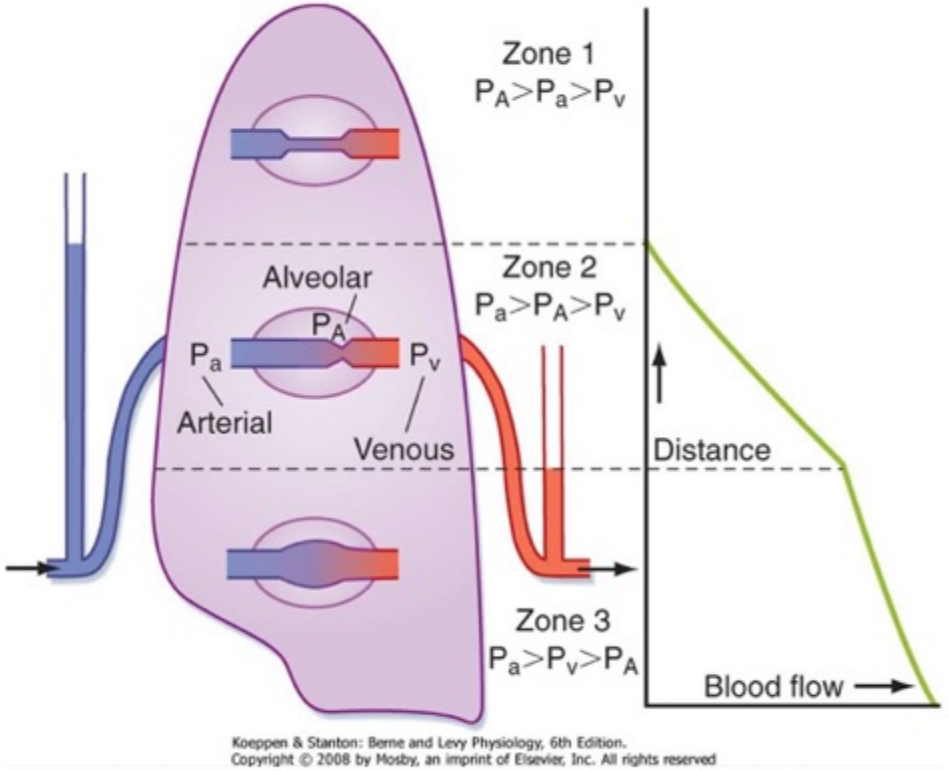
Ventilation in the healthy lung:

- Unevenly distributed
- Gravity causes a distribution of intrapleural pressures
- At normal lung volumes the base of the lung is better ventilated than the apex
- Also influenced by differences in regional mechanics and airflow resistance

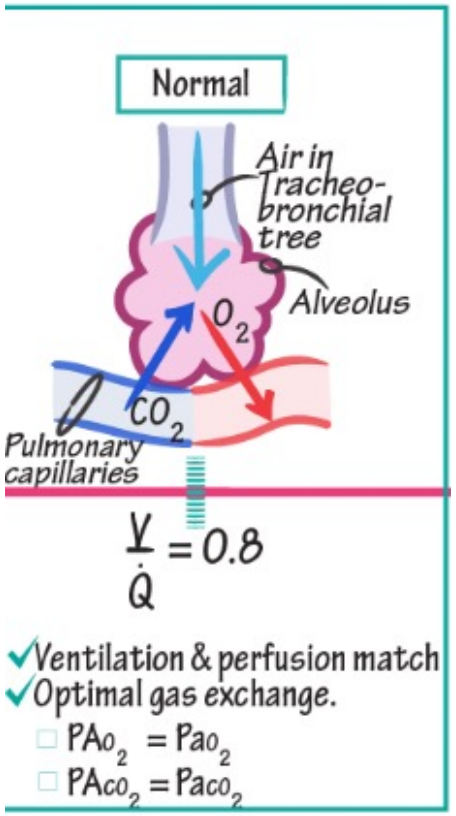


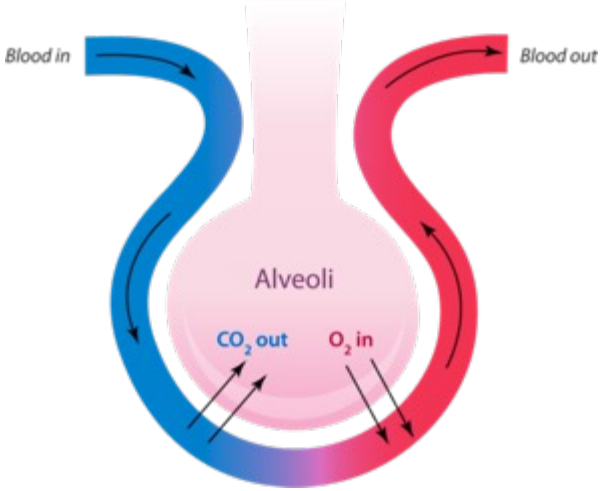
Perfusion in the healthy lung:

- Unevenly distributed
- Explained by hydrostatic pressures within blood vessels
- Also influenced by the arrangement of blood vessels and capillaries at any given level in the lung



Koeppen & Stanton: Berne and Levy Physiology, 6th Edition.
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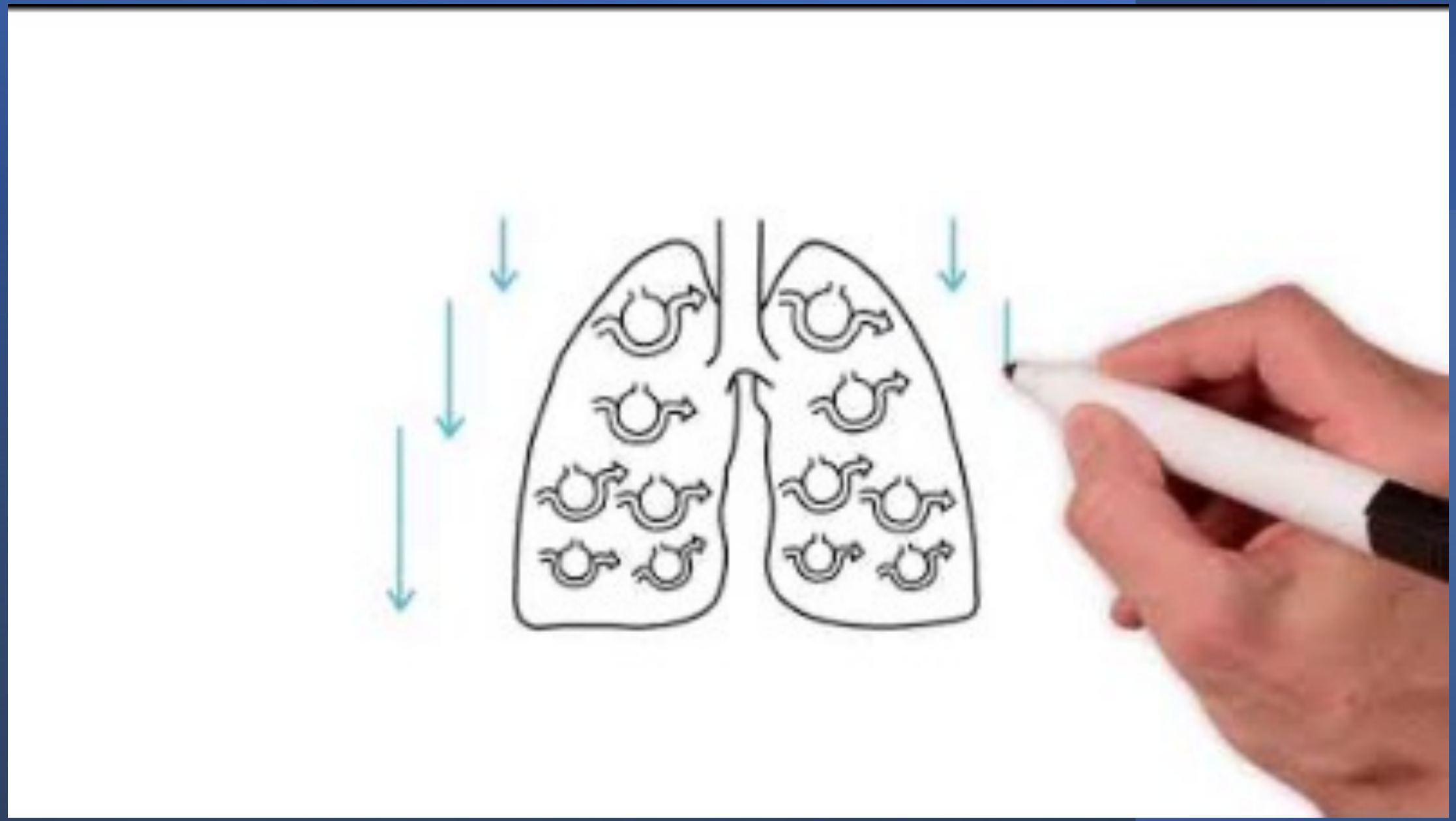


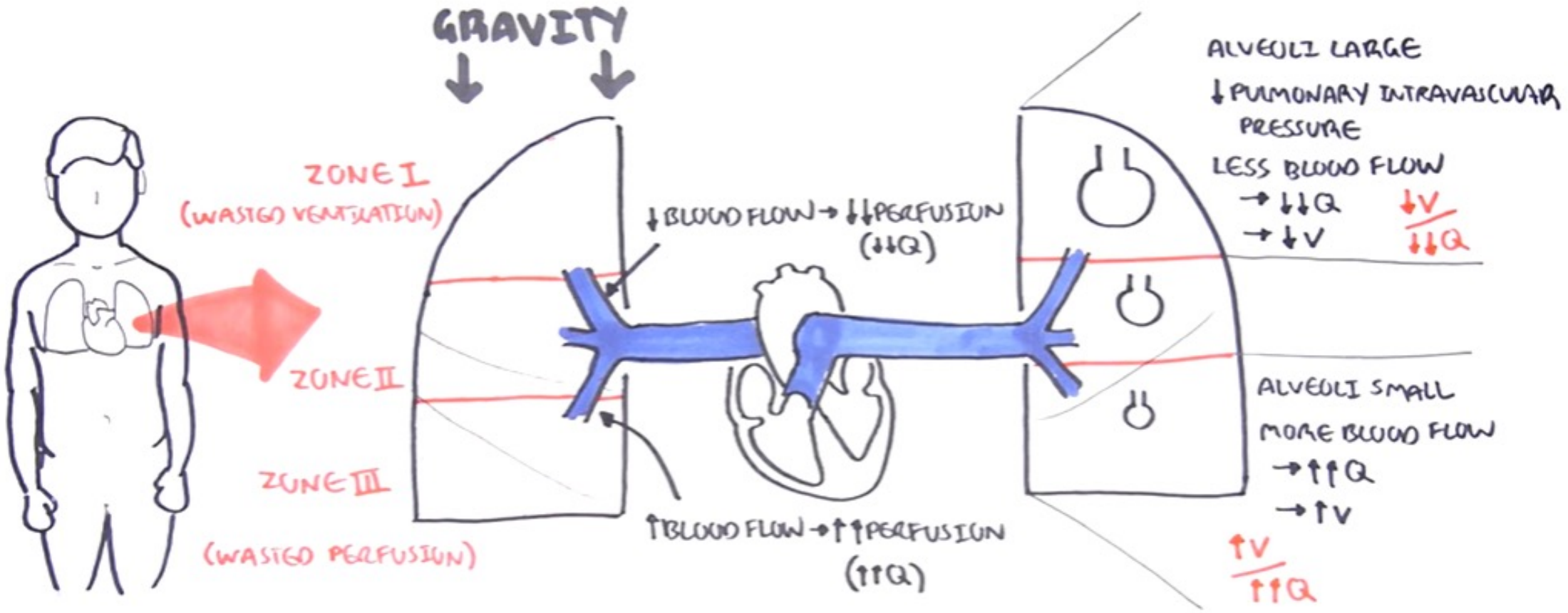
V/Q Matching in Self-Ventilating Adults:

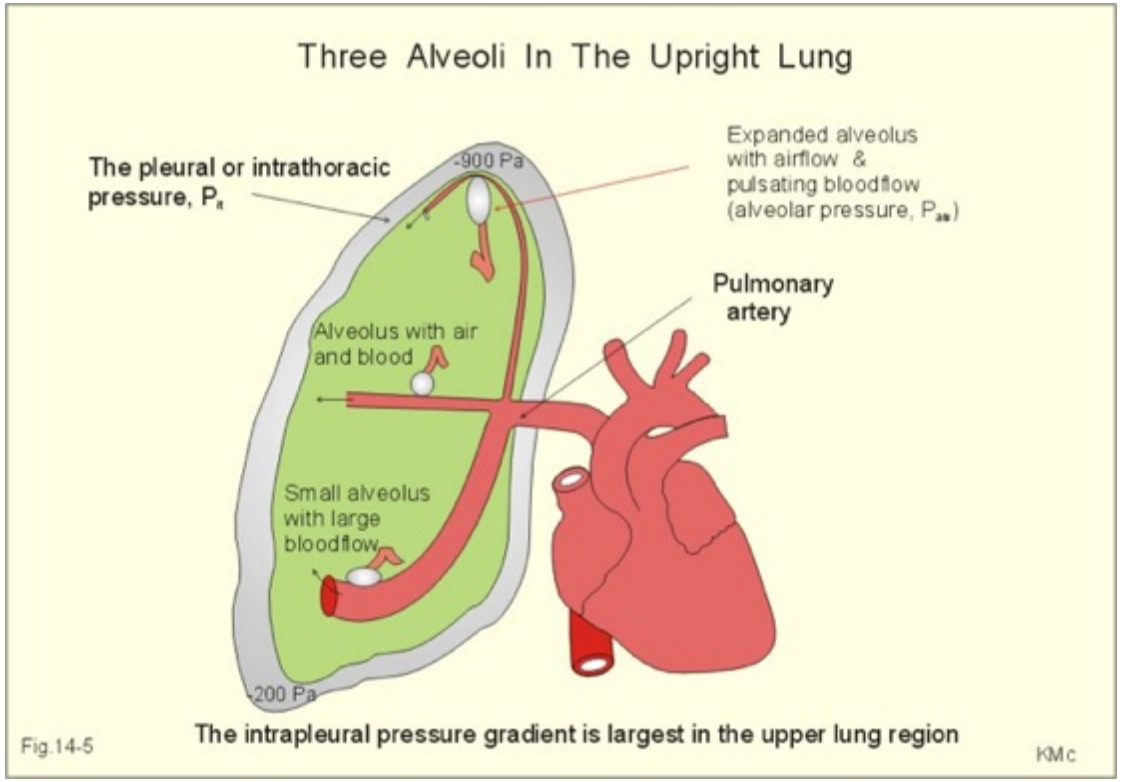
- Optimal V/Q matching occurs in the dependent area of the lungs
- In health, V/Q matching facilitates gas exchange

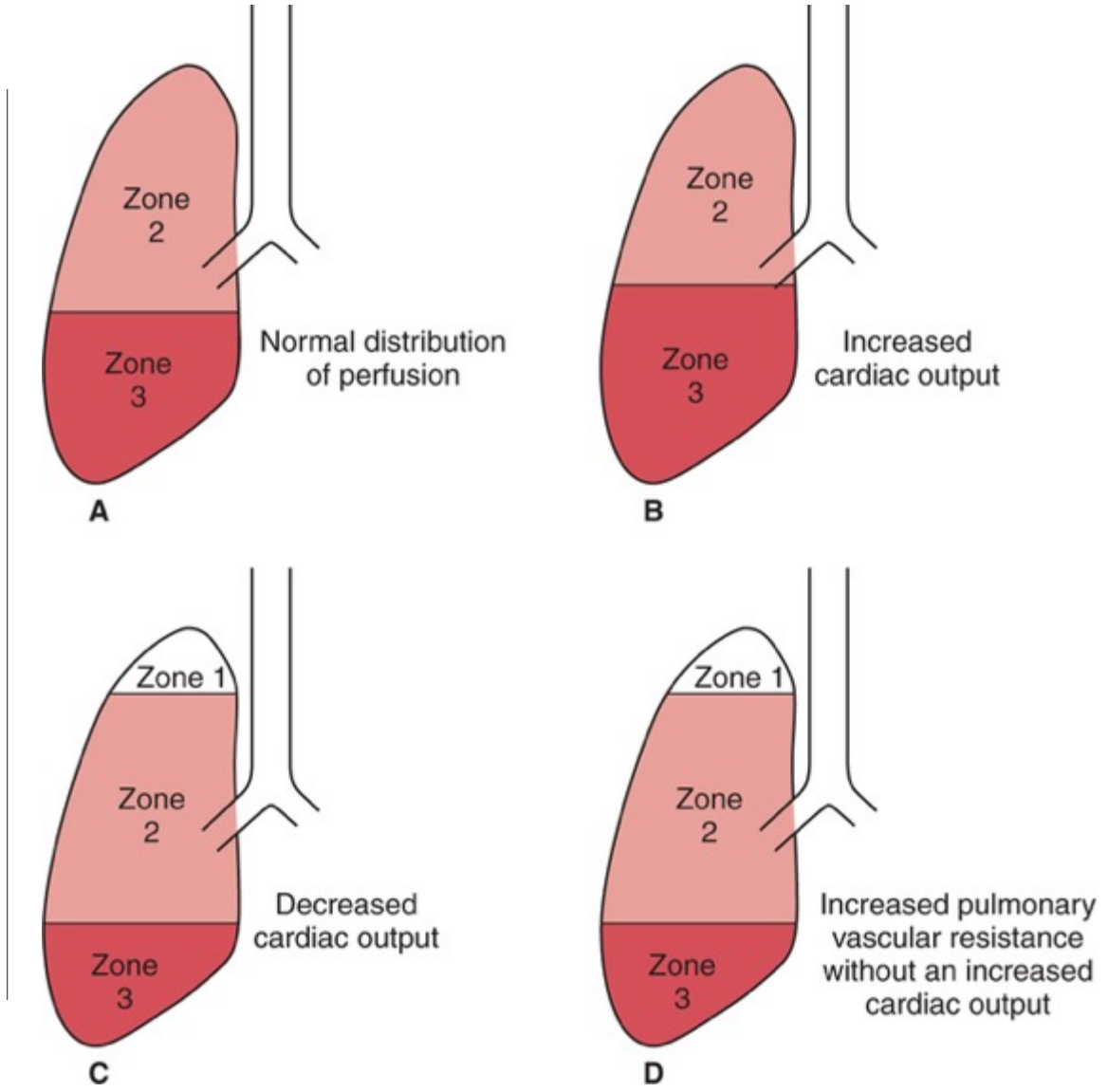
To think about....

- How would you utilise this knowledge to position patients who are hypoxic?

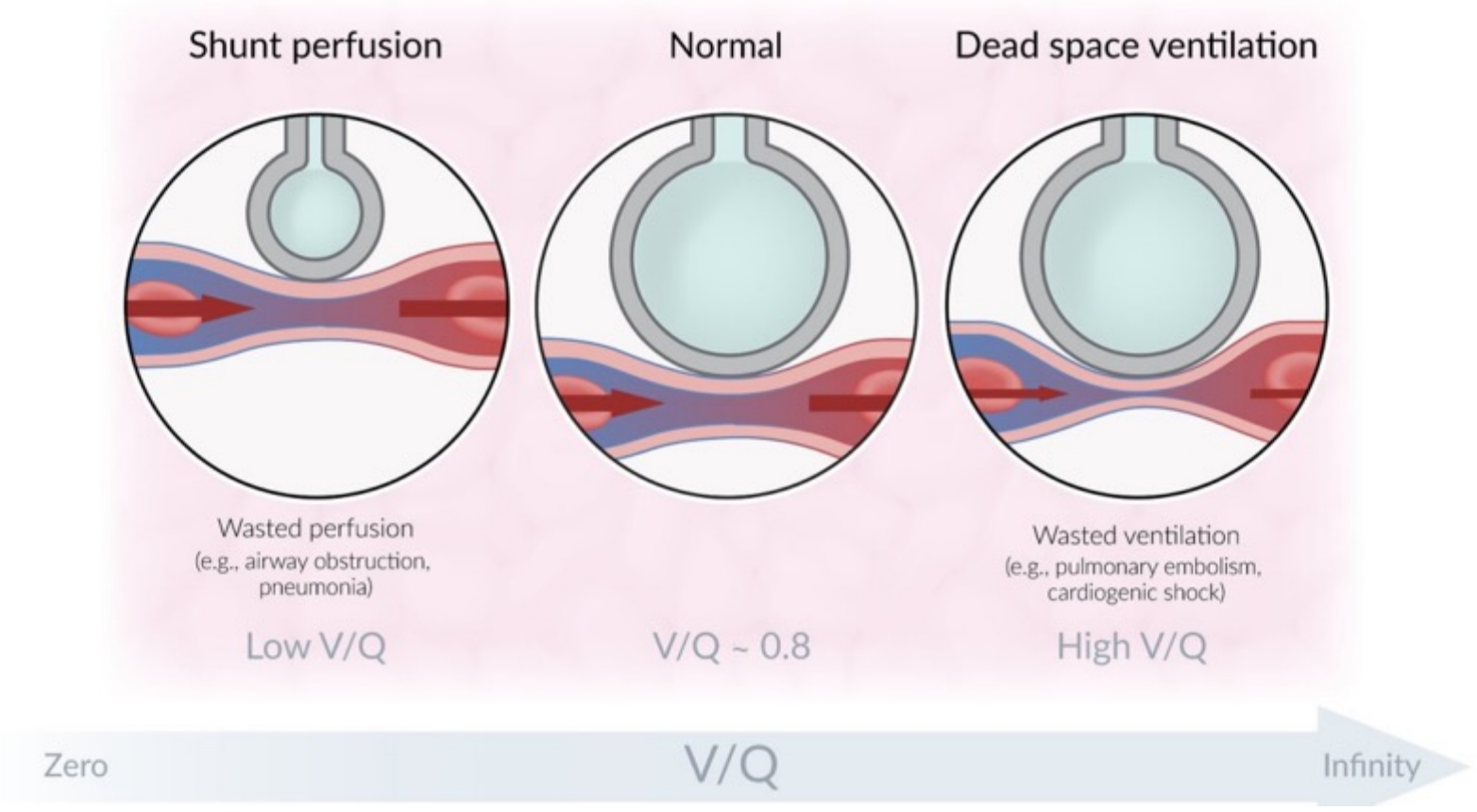



GRAVITY
ZONE I
(WASTED VENTILATION)
ZONE II
ZONE III
(WASTED PERFUSION)
↓BLOOD FLOW → ↓↓PERFUSION
(↓↓Q)
↑BLOOD FLOW → ↑↑PERFUSION
(↑↑Q)
ALVEOLI LARGE
↓PULMONARY INTRAVASCULAR PRESSURE
LESS BLOOD FLOW
→ ↓↓Q
→ ↓V
↓V / ↓↓Q
ALVEOLI SMALL
MORE BLOOD FLOW
→ ↑↑Q
→ ↑V
↑V / ↑↑Q


Three Alveoli In The Upright Lung
The pleural or intrathoracic pressure, P_{ip}
-900 Pa
Expanded alveolus with airflow & pulsating bloodflow (alveolar pressure, P_{alv})
Pulmonary artery
Alveolus with air and blood
Small alveolus with large bloodflow
-200 Pa
Fig.14-5
The intrapleural pressure gradient is largest in the upper lung region
KMc


Zone 2
Zone 3
Normal distribution of perfusion
A
Zone 2
Zone 3
Increased cardiac output
B
Zone 1
Zone 2
Zone 3
Decreased cardiac output
C
Zone 1
Zone 2
Zone 3
Increased pulmonary vascular resistance without an increased cardiac output
D

V/Q Mismatch


Shunt perfusion
Normal
Dead space ventilation
Wasted perfusion
(e.g., airway obstruction, pneumonia)
Wasted ventilation
(e.g., pulmonary embolism, cardiogenic shock)
Low V/Q
V/Q ~ 0.8
High V/Q
Zero
V/Q
Infinity

Maintaining V/Q match

Hypoxic Pulmonary Vasoconstriction

Pulmonary Capillary Recruitment

Clinical Application – Hypoxaemia

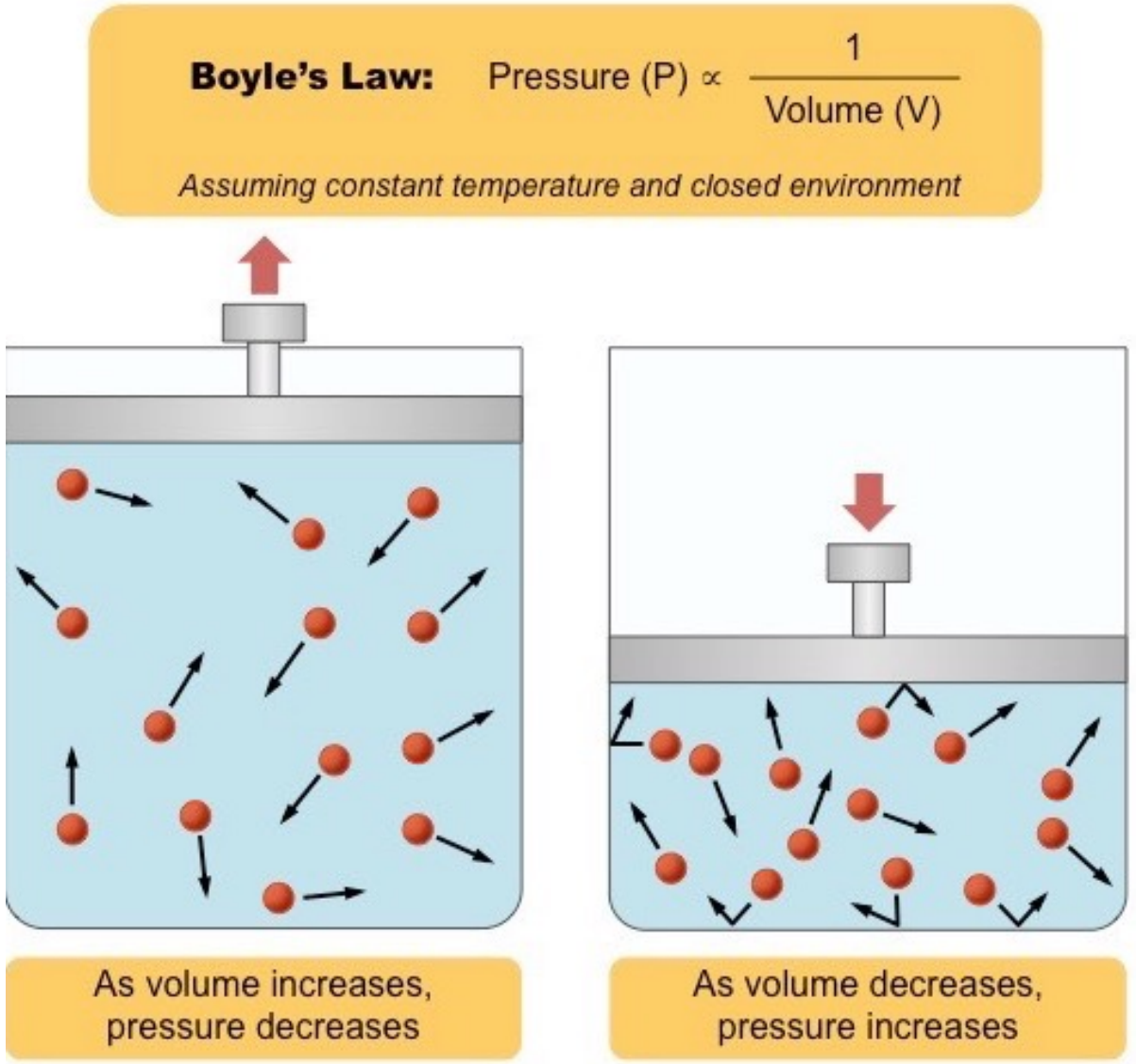
Cause of Hypoxaemia	PAO2	PA-aO2	PaCO2	PaO2 response to increase in FiO2
Diffusion limitation	Normal	Increased	Normal	Improved
Hypoventilation	Reduced	Normal	Increased	Improved
Reduced PiO2	Reduced	Normal		Improved
Low VA/Q	Reduced locally	Increased		Improved
Shunt	Reduced locally	Increased		Minimal Improvement

PAO2: alveolar partial pressure of O2; PA-aO2: alveolar-arterial O2 tension difference; PaCO2: alveolar partial pressure of CO2; PaO2: arterial partial pressure of O2; FiO2: inspiratory O2 fraction; PiO2: partial pressure of O2 in inspired gas; VA/Q: rate of alveolar ventilation to perfusion

Mechanics of Ventilation

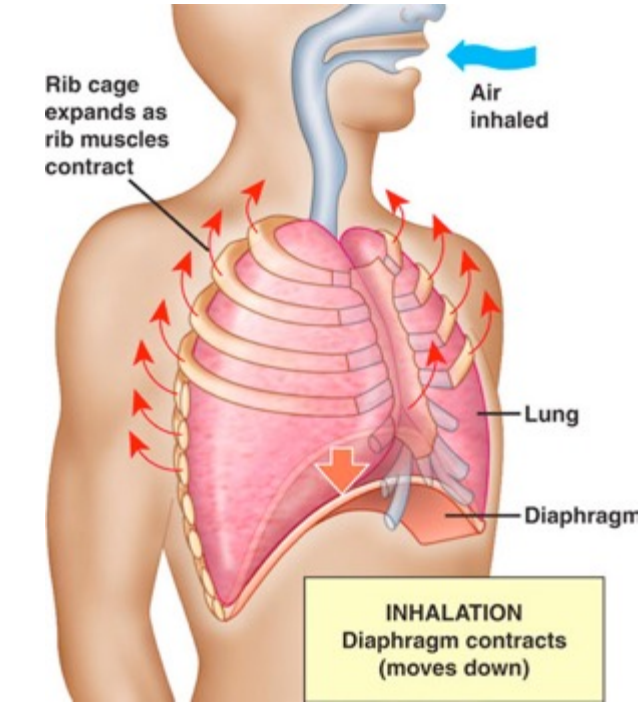
BOYLES LAW:

- Changes in intrapulmonary pressure occur as result of changes in lung volume
  - Pressure of gas is inversely proportional to volume
- Increase in lung volume = decreased intrapulmonary pressure
  - Air goes in
- Decrease in lung volume raises intrapulmonary pressure above atmosphere
  - Air goes out



Mechanics of Breathing: Inhalation

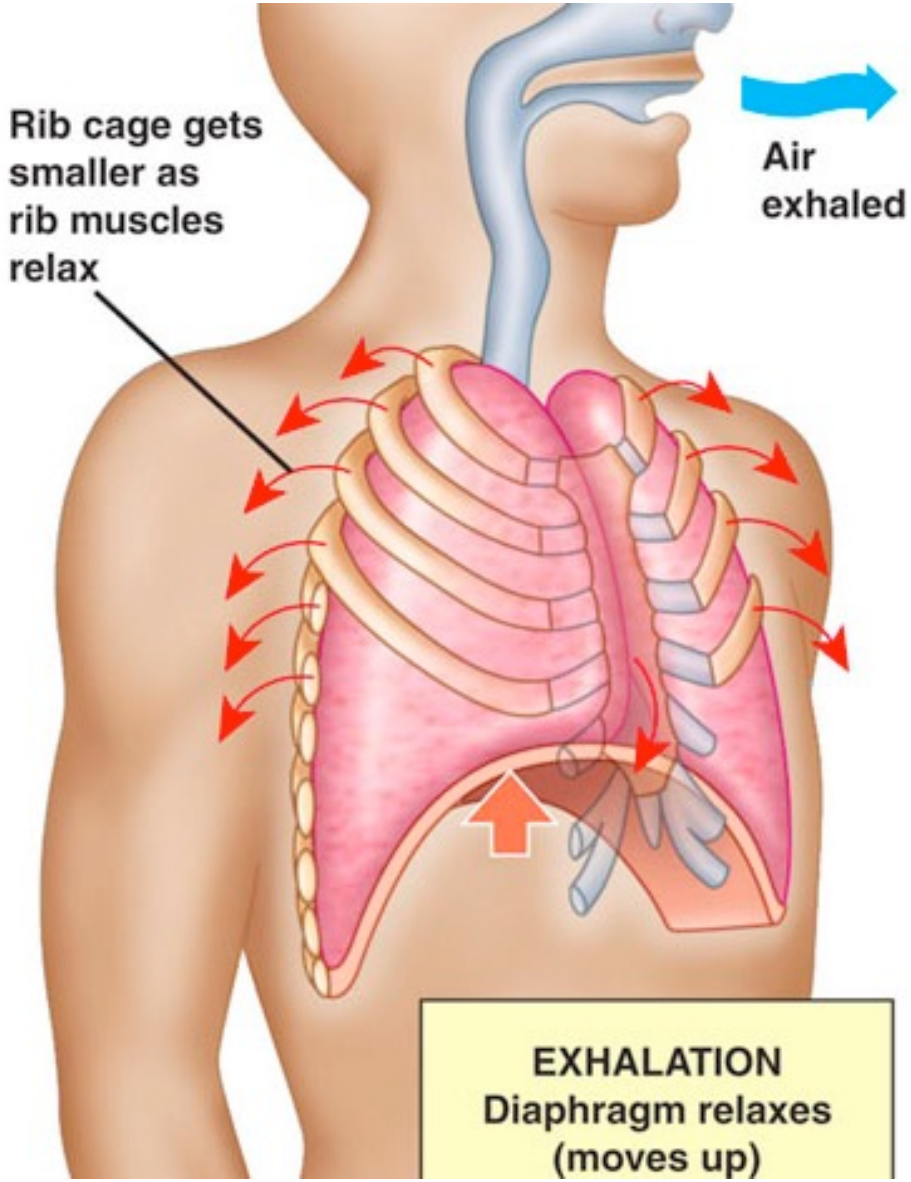
- Diaphragm contracts and flattens
- External intercostal muscles draw the ribs upward and outward
- Increase in thoracic volume
- Decrease in intrapulmonary pressure
- Active process

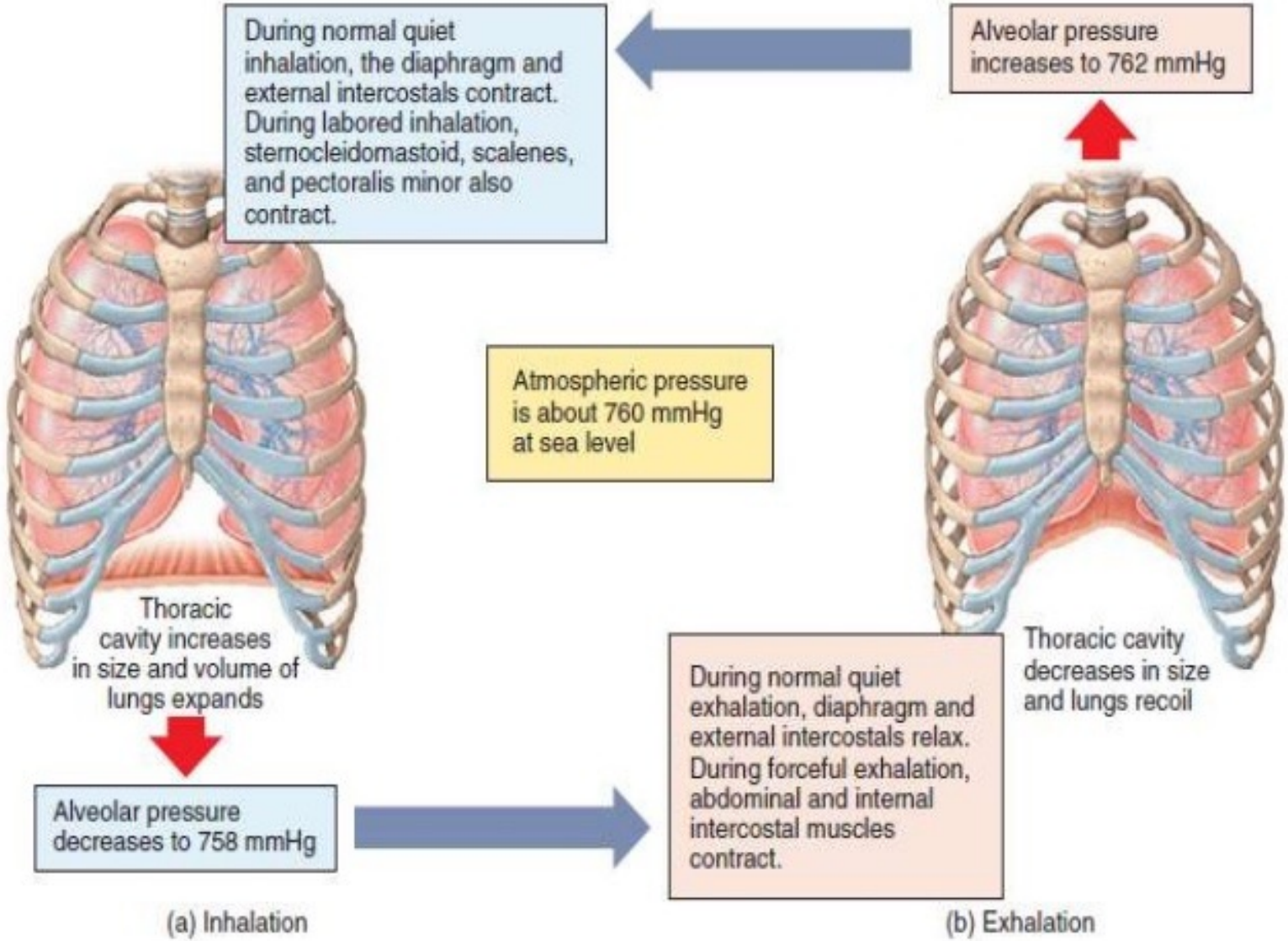




Mechanics of Breathing: Exhalation

- Diaphragm and external intercostal muscles relax
- Thoracic cavity volume decreases
- Increased intrapulmonary pressure - exceeds the atmospheric pressure
- Air is expelled from the lungs
- Passive process



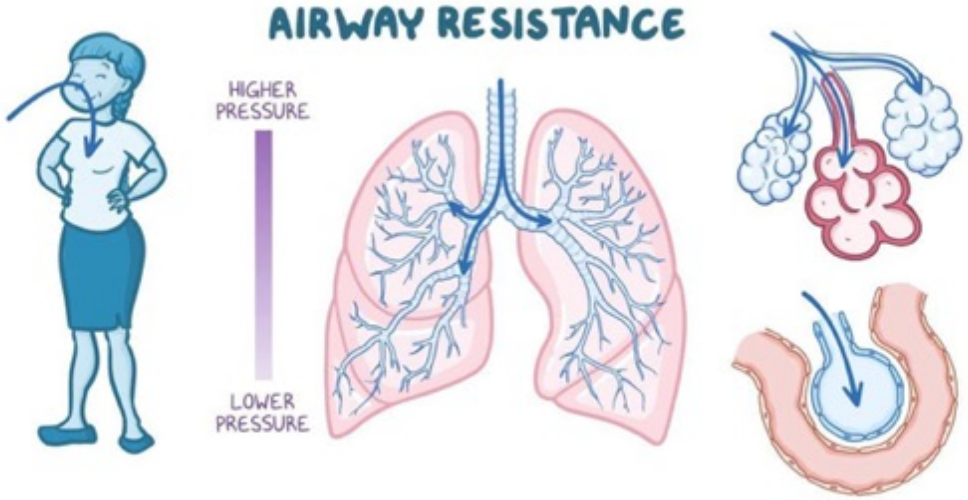

During normal quiet inhalation, the diaphragm and external intercostals contract. During labored inhalation, sternocleidomastoid, scalenes, and pectoralis minor also contract.
Alveolar pressure increases to 762 mmHg
Atmospheric pressure is about 760 mmHg at sea level
Thoracic cavity increases in size and volume of lungs expands
Alveolar pressure decreases to 758 mmHg
During normal quiet exhalation, diaphragm and external intercostals relax. During forceful exhalation, abdominal and internal intercostal muscles contract.
Thoracic cavity decreases in size and lungs recoil
(a) Inhalation
(b) Exhalation

PHYSICAL FACTORS affecting ventilation - RESISTANCE:

- = Force that must be overcome during breathing

- **Airflow Resistance**
  - Caused by friction
  - Depends on speed of airflow & calibre of airway

- **Elastic (Elastance) Resistance**
  - Contributed to by lung tissue, alveolar surface liquid and chest wall



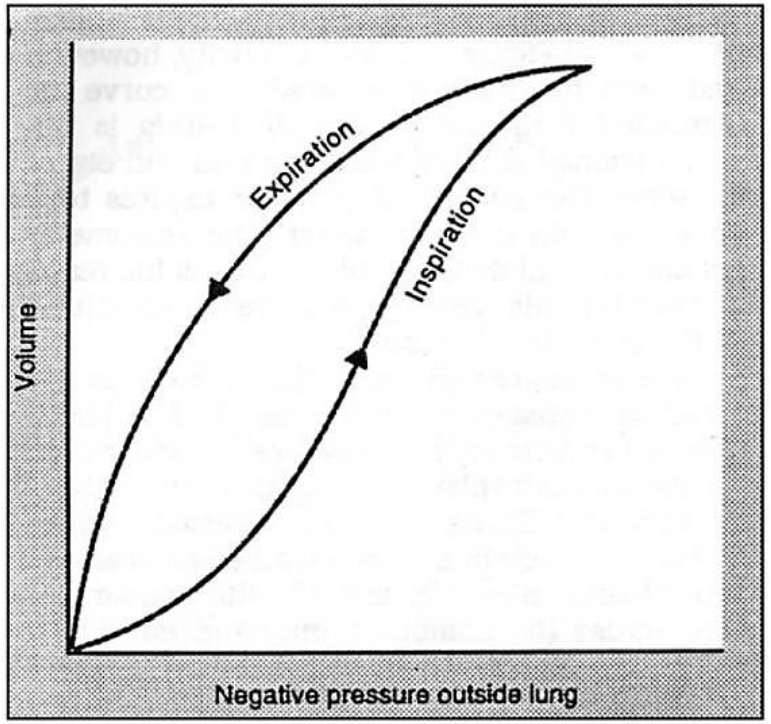
What could increase resistance?

PHYSICAL FACTORS affecting ventilation – COMPLIANCE:

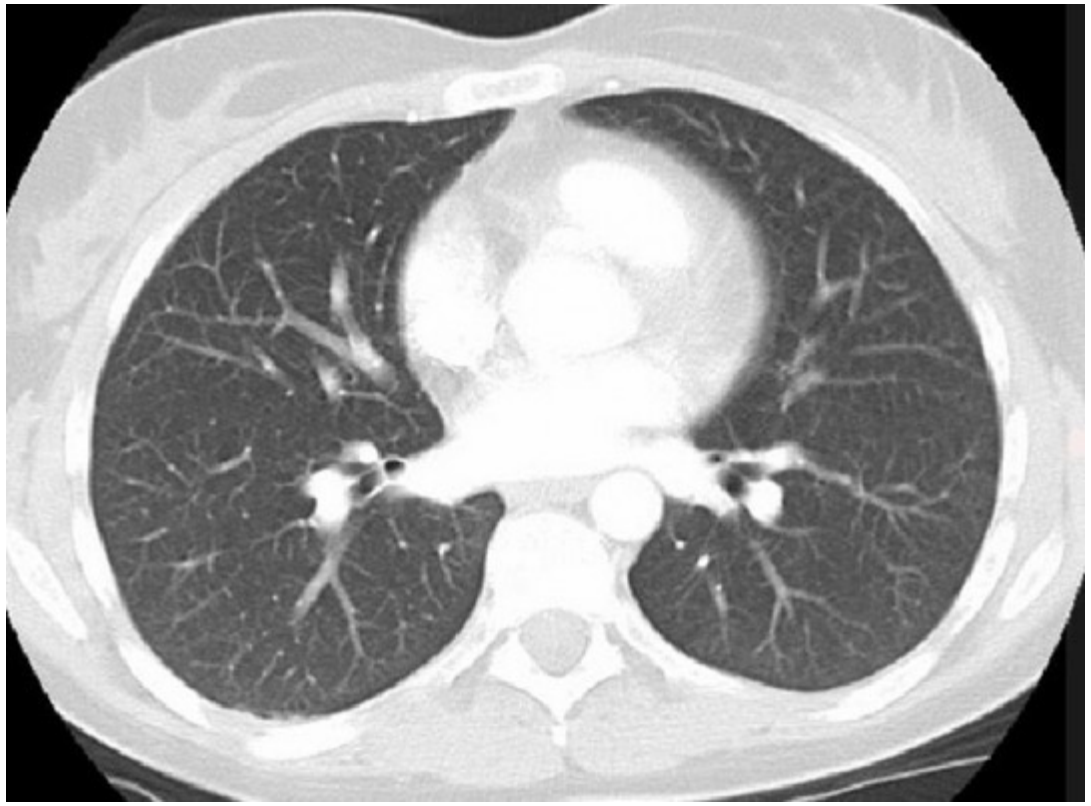
- Willingness of the lungs to distend

- Elastance – willingness of the lungs to return to resting state
- Compliance is inversely proportional to elastance

- Relationship between volume & pressure
  - eg: how much pressure (WOB) is required to expand the lungs?
  - Compliance =
    - change in volume / change in pressure



Low lung compliance (stiff lungs): Lung CT scan- Pulmonary fibrosis

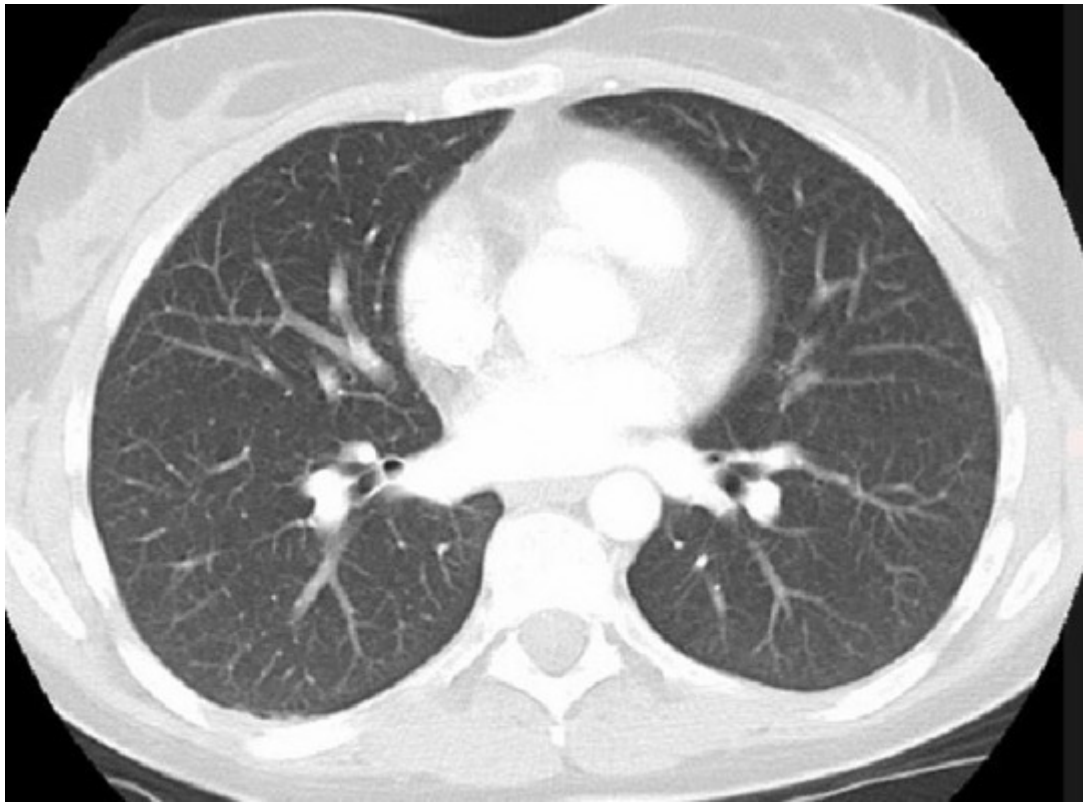


NORMAL

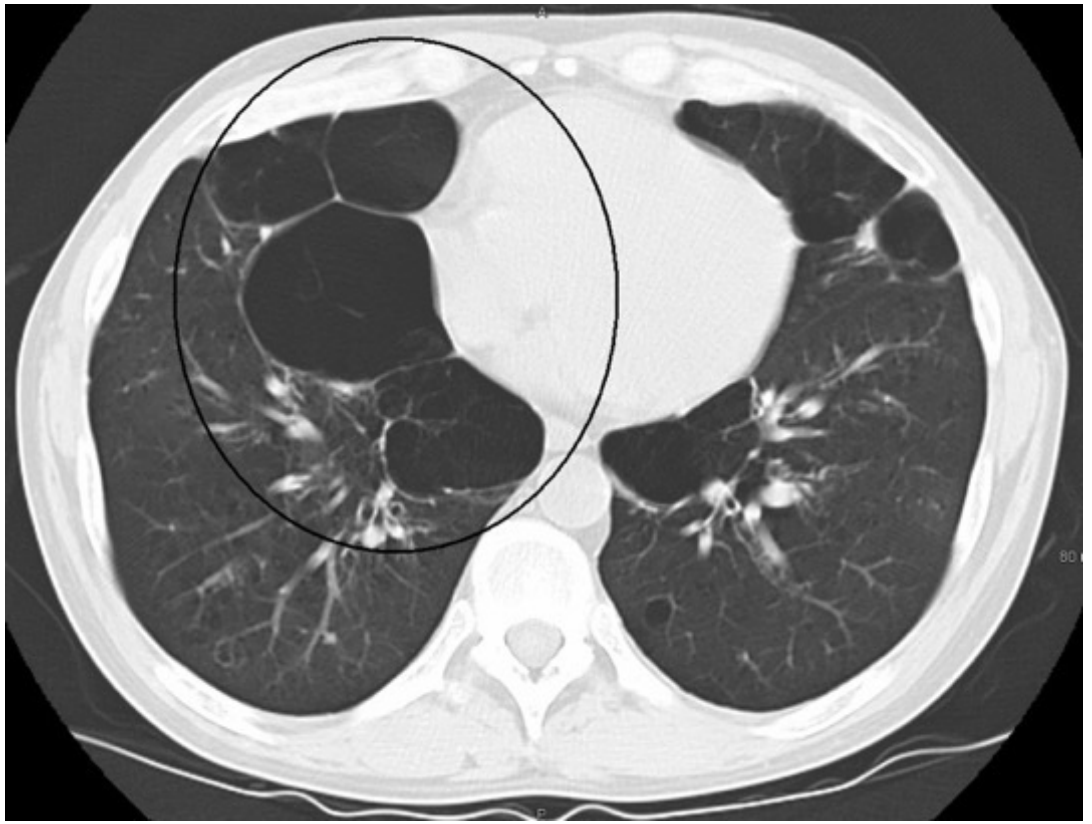


PULMONARY FIBROSIS

High lung compliance (floppy airways): Lung CT scan- Emphysema



NORMAL



EMPHYSEMA

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